

CptS 570 Machine Learning Midterm Exam, Fall 2020

Exam date: Oct 29

Instructions.

- The maximum score of the exam is 100 points.
- Read all the questions before starting to answer. Try to answer those questions, which you think are easy from your perspective first.
- Work efficiently. Most questions don't require much work. If you are spending more than 5 mins on any question, then you should try to re-think about it.
- Keep your answers short and simple.

- **Short Questions (35 points)**

Please keep your answers short (one sentence). For the True / False questions, please also provide a short justification.

0. **(2 points)** Incredibly hard question! What is the course number for this machine learning class?

1. **(3 points)** The training error of K-NN (nearest neighbor) classifier is ZERO. (True/False)

2. **(5 points)** Logistic regression classifier learns a non-linear decision boundary because the logistic (sigmoid) function is non-linear. (True/False)

3. **(5 points)** The greedy decision tree learning algorithm based on information gain is guaranteed to find the optimal decision tree (True / False)

4. **(5 points)** If you are given m data points, and use half for training and half for testing, the difference between training error and test error decreases as m increases. (True/False)

Please write one sentence justification

5. **(4 points)** The error of a classifier measured over its training set provides a optimistically biased estimate of the true error of the hypothesis. (True/False)

6. **(4 points)** Why does the kernel trick allow us to solve SVMs with high dimensional feature spaces, without significantly increasing the running time?

7. **(3 points)** Passive learning is label-efficient when compared to active learning (True/False)

8. **(4 points)** Assume our training data contains exactly one training example. What is the average log-likelihood on this training data?

a) $-\infty$

b) -1

c) $-1/2$

d) 0

e) $1/2$

f) 1

g) ∞

• **Parameter Estimation (40 points)**

9. **(5 points)** Suppose we have a training set that is linearly separable with margin $\gamma > 1$. If we run both Passive-Aggressive (PA) and Perceptron classifier learning algorithms, will we get the same classifier? Please explain your reasoning.

10. **(5 points)** Suppose D is the data and θ stands for the parameters. Which of the following corresponds to maximum likelihood hypothesis?

$$\arg \max_{\theta} P(D|\theta) \tag{1}$$

$$\arg \max_{\theta} P(\theta|D) \tag{2}$$

11. **(5 points)** If a data point y follows the Poisson distribution with rate parameter θ , then the probability of a single observation y is

$$p(y|\theta) = \frac{\theta^y e^{-\theta}}{y!}, \text{ for } y = 0, 1, 2, \dots \tag{3}$$

You are given data points $y_1, y_2, \dots, y_n$ independently drawn from a Poisson distribution with parameter θ . Write down the log-likelihood of the data as a function of θ .

12. **(10 points)** Suppose we have a training data with classes $Y \in \{0, 1\}$ and d features with each feature $f_i \in \{+1, -1\}$. To improve the performance of the binary classifier, Jana decided to duplicate each feature. Each training example now has $2d$ features with $f_{d+i} = f_i$ for $i = 1, 2, \dots, d$. How does the training problem with *original* feature set and *double* feature set compare? Assume that there are same number of training examples for both positive and negative class, and in case of ties, you will chose positive class.

For a Perceptron classifier, select all that apply.

- a) Test accuracy with original feature set could be higher
- b) Test accuracy with double feature set could be higher
- c) Test accuracy will be same with both original and double feature set
- d) Can't say for sure

Please write one sentence justification

For a Naive Bayes classifier, select all that apply.

- a) Test accuracy with original feature set could be higher
- b) Test accuracy with double feature set could be higher
- c) Test accuracy will be same with both original and double feature set
- d) On a given training example, the conditional probability $P(Y|f_1, \dots)$ will could be more extreme (close to 0 or 1) with original feature set
- e) On a given training example, the conditional probability $P(Y|f_1, \dots)$ will could be more extreme (close to 0 or 1) with double feature set

Please write one sentence justification

13. **(7 points)** Jana has to decide between the standard Perceptron algorithm: works directly with $\phi(x)$; and the kernelized perceptron algorithm: works with the kernel function K and does not explicitly compute $\phi(x)$. Suppose m is the total number of training examples and k is the total number of features in mapping $\phi(x)$. When should Jana prefer the kernelized perceptron over standard perceptron algorithm based on the criteria of time complexity and space complexity.

- a) $k \gg m$
- b) $k \ll m$
- c) always
- d) never

Please write your justification

14. **(8 points)** Suppose we are training a Dual Perceptron for a three-class (*good*, *bad*, *ugly*) problem. There are four training examples x_1, x_2, x_3, x_4 . The dual weights are currently.

$\alpha_{good} = (-1, -1, -1, -1)$ for class *good*

$\alpha_{bad} = (-1, +1, +1, -1)$ for class *bad*

$\alpha_{ugly} = (-1, -1, -1, -1)$ for class *ugly*

Consider the fourth training example x_4 with correct label *good* and following kernel evaluations:

$$K(x_1, x_4) = 1; K(x_2, x_4) = 2; K(x_3, x_4) = 1; K(x_4, x_4) = 3$$

- a) Which classification label is predicted for the fourth training example x_4 with the current dual weights?

b) What are the dual weights $(\alpha_{good}, \alpha_{bad}, \alpha_{ugly})$ after the update that incorporates the fourth training example?

• **Decision Trees, Ensemble Methods, and Misc. (25 points)**

15. **(5 points)** Suppose we want to learn a Bayes classifier from three categorical features f_1, f_2, f_3 that can take four, five, and six values respectively; and one categorical output y (seven classes). Answer the following:

1) How many parameters must be estimated to train a naive Bayes classifier?

2) How many parameters do we need to estimate if we do **NOT** make the naive Bayes conditional independence assumption?

(3 points) Consider training a decision tree using the following data. Let f_1, f_2 , and f_3 denote the three features.

Table 1: Training data for decision tree learning.

f_1	f_2	f_3	Class label
6	3	1	+1
2	7	1	-1
9	6	1	+1
4	2	1	-1

Which of the following splits at the root node gives the highest information gain?

- a) $f_1 > 2$
- b) $f_2 > 3$
- c) $f_1 > 4$
- d) $f_2 > 6$

16. **(4 points)** Which of the following are true for Bagging?

- a) In bagging, we choose random subsamples of the input training examples with replacement
- b) In bagging, we choose random subsamples of the input training examples without replacement
- c) Bagging is ineffective with perceptron, because all of the learners learn exactly the same decision boundary
- d) Bagging only works with decision trees
- e) Bagging only works when the classifiers in the ensemble have diversity

17. **(3 points)** Please state the Occam's Razor principle and explain it using decision tree as a classifier.

18. **(3 points)** If the training set contains noise in class labels, which ensemble learning method among bagging and boosting do you expect to hurt more by the label noise? Why?

19. **(3 points)** How will you detect over-fitting and under-fitting using the learning curves?

20. **(3 points)** What is the main difference between *Maximum Likelihood Estimation (MLE)* and *Maximum A Posteriori (MAP)* estimation of parameters θ from given data D ?

Extra Sheet

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